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GRADE A+
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Retail company Sales

Submitted By:

Name: Kavivarshini S

Reg. No.: 2411021240004

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Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

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Retail company Sales

Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in multiple CSV files, customer data is stored separately, and product information is available through CSV and MySQL. Management wants a centralized reporting system. The objective is to connect these sources to Power BI, prepare the data, create relationships, and develop an interactive dashboard for business analysis.

Step 1: Data Collection

- Obtained Sales Data for **2020, 2021, and 2022** from CSV files.
- Obtained **Customer Lookup** data containing customer information.
- Obtained **Product Lookup** data containing product details.
- Used the **AdventureWorks MySQL database** as an additional data source.

Step 2: Data Import

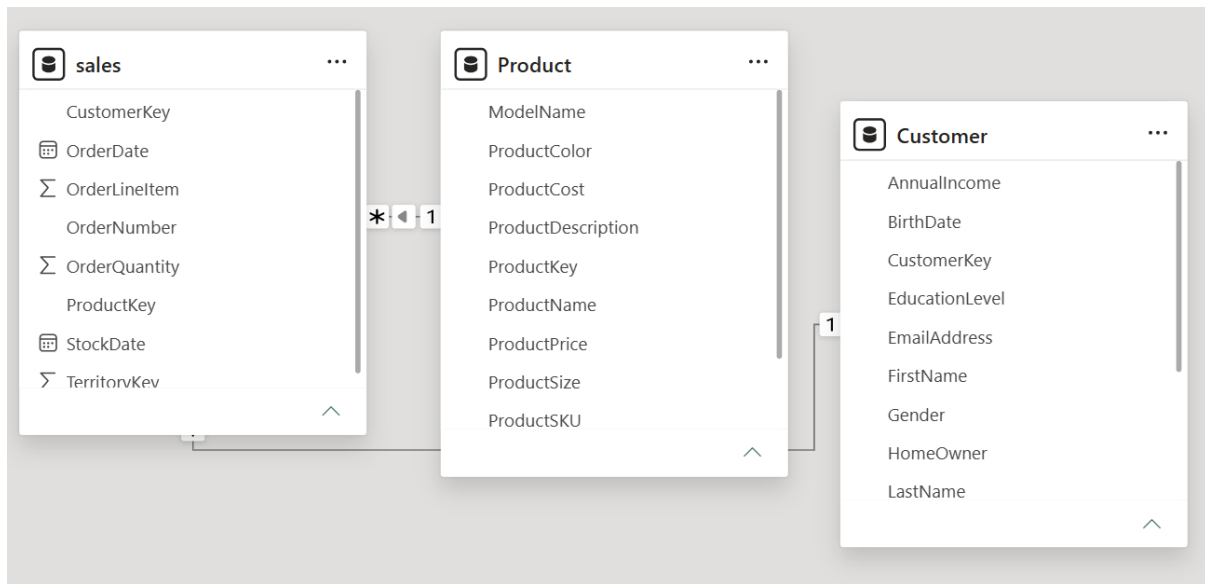
- Opened **Power BI Desktop**.
- Imported the Sales, Customer, and Product datasets using the **Get Data** option.
- Connected Power BI to the **MySQL database**.
- Loaded the required tables into Power BI.

Step 3: Data Cleaning and Transformation

- Opened the datasets using **Power Query Editor**.
- Combined the 2020, 2021, and 2022 Sales files using **Append Queries**.
- Converted **OrderDate** and **StockDate** into proper Date format.
- Corrected incorrect data types such as CustomerKey, ProductKey, and OrderQuantity.
- Removed invalid or unwanted records from the Customer table.
- Verified the cleaned data and prepared it for modeling.

Step 4: Data Modeling

- Created a relationship between the **Sales and Product** tables using ProductKey.
- Created a relationship between the **Sales and Customer** tables using CustomerKey.
- Verified the relationship cardinality and filtering direction.
- Created a **TotalSales DAX measure** using OrderQuantity and ProductPrice.



Step 5: Dashboard Development

- Created KPI cards for **Total Orders**, **Total Products**, **Total Customers**, and **Total Sales**.
- Created a bar chart for **Sales by Product Subcategory**.
- Created a bar chart for **Sales by Territory/Region**.
- Created a line chart for **Monthly Sales Trend**.
- Added slicers for **Order Date**, **Product Subcategory**, and **Territory**.
- Arranged the visuals to create an interactive business dashboard.



Output

The final Power BI dashboard provides:

- **Total Sales Performance**
- **Total Orders and Customers**
- **Product and Category-wise Sales Analysis**
- **Regional/Territory Sales Performance**
- **Monthly Sales Trends**
- **Interactive Filters and Slicers**

The dashboard provides a centralized view of retail sales performance and allows users to analyze the data based on different products, territories, and time periods.

Conclusion

This project successfully integrated **Sales, Customer, Product, and MySQL data sources** into Power BI. The data was cleaned, transformed, and connected using appropriate relationships. DAX was used to calculate Total Sales, while KPI cards, bar charts, line charts, and slicers were used to create an interactive dashboard. The final dashboard helps users understand sales performance and supports **data-driven business decision-making**.

Learning Outcomes

Through this project, I learned:

- How to import and combine multiple **CSV datasets** into Power BI.
- How to connect **Power BI with MySQL**.
- How to clean and transform data using **Power Query**.
- How to change and verify appropriate **data types**.
- How to append multiple datasets into a single Sales table.
- How to create relationships between **Sales, Product, and Customer** tables.
- How to create **KPI cards, bar charts, and line charts**.
- How to use **slicers and filters** for interactive analysis.
- How to design a dashboard that converts raw business data into meaningful insights

GitHub link : <https://github.com/Kavivarshini13/Business-Analytics->